

ACROSS TIME AND SPACE

VISUALIZING ENVIRONMENTAL CHANGE FROM MULTI-TEMPORAL SATELLITE IMAGE

01 Pipeline

READING IMAGE

- Sort multi-temporal images by time
- Call multi-image analysis (>2)



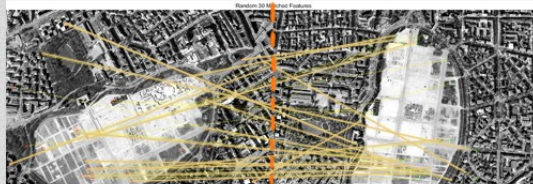
PREPROCESSING

- Pairwise comparison with the first image
- Grayscale adjustment



MATCHING

- detectSURFfeatures: feature matching
- estgeform2d: mapping relationship
- imwarp: image warping



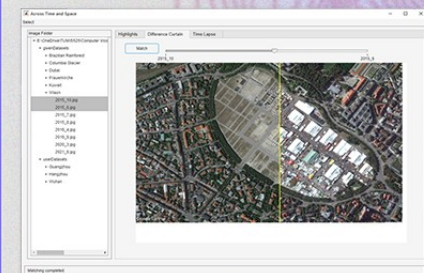
RENDERING

- Align with the first image



02 GUI

THREE VIEWING MODES



04 METHODS

Adjust parameter thresholds

Use stable and efficient SURF library

Always compare with the first image

Image subdivision into 4 or 9 parts

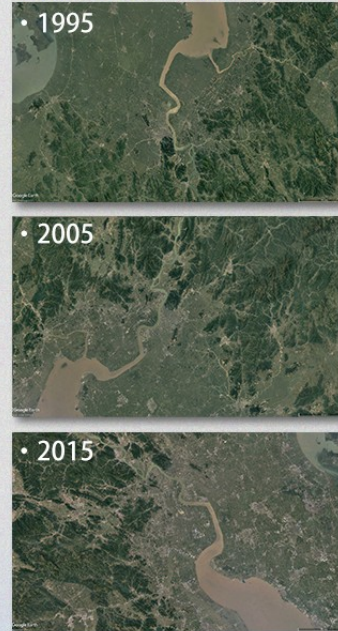
Fill black regions

05 RESULTS



Hangzhou, China

ORIGINAL IMAGE



OUTPUT IMAGE



THE 1ST

STANDARD

OUTPUT IMAGE

1985

2020